

```
[1] "Mazda RX4"      "Mazda RX4 Wag"   "Datsun 710"
[4] "Hornet 4 Drive"  "Hornet Sportabout" "Valiant"

> mpg <- c(21.0, 21.0, 22.8, 21.4, 18.7, 18.1)
> mpg
[1] 21.0 21.0 22.8 21.4 18.7 18.1
```

```
> cyl <- c(6, 6, 4, 6, 8, 6)
> cyl
[1] 6 6 4 6 8 6
```

```
> disp <- c(160, 160, 108, 258, 360, 225)
> disp
[1] 160 160 108 258 360 225
```

```
> hp <- c(110, 110, 93, 110, 175, 105)
> hp
[1] 110 110 93 110 175 105
```

```
> wt <- c(2.620, 2.875, 2.320, 3.215, 3.440, 3.460)
> wt
[1] 2.620 2.875 2.320 3.215 3.440 3.460
```

```
> gear <- c(4, 4, 4, 3, 3, 3)
> gear
[1] 4 4 4 3 3 3
```

```
> cars <- data.frame(name, mpg, cyl, disp, hp, wt, gear)
> cars
      name mpg cyl disp hp wt gear
1  Mazda RX4 21.0  6 160 110 2.620  4
2  Mazda RX4 Wag 21.0  6 160 110 2.875  4
4  Hornet 4 Drive 21.4  6 258 110 3.215  3
3   Datsun 710 22.8  4 108  93 2.320  4
5 Hornet Sportabout 18.7  8 360 175 3.440  3
6    Valiant 18.1  6 225 105 3.460  3
>
```

The type of 'cars' is a list.

```
> typeof(cars)
[1] "list"
```

The structure of the data frame can be seen using the `str()` function, as illustrated below:

```
> str(cars)
'data.frame':  6 obs. of  7 variables:
 $ name: chr  "Mazda RX4" "Mazda RX4 Wag" "Datsun 710"
"Hornet 4 Drive" ...
```

```
$ mpg : num  21 21 22.8 21.4 18.7 18.1
$ cyl : num   6 6 4 6 8 6
$ disp: num  160 160 108 258 360 225
$ hp  : num  110 110 93 110 175 105
$ wt  : num   2.62 2.88 2.32 3.21 3.44 ...
$ gear: num   4 4 4 3 3 3
```

You can retrieve the individual values using indices for the data frame. You can also specify a range for the indices, as shown below:

```
> cars[1:2, 2]
[1] 21 21
```

The column values can be obtained by mentioning the column name in the data frame. For example:

```
> cars[, "wt"]
[1] 2.620 2.875 2.320 3.215 3.440 3.460
```

```
> cars$wt
[1] 2.620 2.875 2.320 3.215 3.440 3.460
```

You can also retrieve a section of the data that satisfies a condition using the `subset()` function, as illustrated below:

```
> subset(cars, subset = hp > 100)
      name mpg cyl disp hp wt gear
1  Mazda RX4 21.0  6 160 110 2.620  4
2  Mazda RX4 Wag 21.0  6 160 110 2.875  4
4  Hornet 4 Drive 21.4  6 258 110 3.215  3
5 Hornet Sportabout 18.7  8 360 175 3.440  3
6    Valiant 18.1  6 225 105 3.460  3
```

The cars can be sorted based on the displacement data in ascending order using the `order()` function, as shown below:

```
> order(cars$disp)
[1] 3 1 2 6 4 5
```

You are encouraged to try the above data structures and their supported functions on your data to understand their use and functionality. In the next part, we shall learn how to import data from various sources into R.  

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